Protein PP Set 1

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# I. Amino Acids: Properties & Classification

**1. Which of the following amino acid have nonpolar side chain?** (Biochemistry Midterm 2021)

**Answer:** Phenylalanine
**Explanation:** Lippincott classifies Phenylalanine as having an aromatic, nonpolar side chain that promotes hydrophobic interactions.

**2. Which among the following is a non-essential amino acid?** (Biochemistry Mids 2020)

**Answer:** Serine
**Explanation:** Serine can be synthesized from 3-phosphoglycerate (an intermediate of glycolysis), whereas essential amino acids like Histidine and Lysine cannot be synthesized *de novo* by humans.

**3. Which amino acid is classified as essential?** (Biochem 2025 Mock DDC)

**Answer:** Valine

**4. Sulphur containing amino acids are?** (Biochemistry Mids 2020)

**Answer:** Cysteine and Methionine

**5. Aromatic amino acids include which of the following?** (Biochemistry Mids 2020)

**Answer:** Phenylalanine, Tyrosine, and Tryptophan

**6. Which is the smallest amino acid?** (Biochem Term Paper Dikiohs)

**Answer:** Glycine **Explanation:** Its side chain is a single hydrogen atom, allowing it to fit into tight places where other amino acids cannot.

**7. Which amino acid contains a thiol (-SH) group?** (Biochem Proff 2025)

**Answer:** Cysteine

**8. What is the average molecular weight of an amino acid residue in a protein?** (Biochem 2025 Mock DDC)

**Answer:** 110 Da

**9. The peptide bond is formed between which two groups?** (Biochem 2025 Mock DDC)

**Answer:** $\alpha$-amino group and $\alpha$-carboxyl group

**10. Which amino acid is a precursor of a mediator of allergies and inflammation (Histamine)?** (Biochemistry Midterm 2021)

**Answer:** Histidine

**11. Which amino acid is most compatible with an $\alpha$-helical structure?** (Biochemistry Midterm 2021)

**Answer:** Alanine **Explanation:** Alanine has a small, uncharged side chain that does not interfere with the spiral of the helix.

**12. Which amino acid is a precursor for the synthesis of Thyroid hormones (T3, T4)?** (Biochem Proff 2025)

**Answer:** Tyrosine

**13. Which amino acid is a precursor for Niacin (Vitamin B3)?** (Biochemistry Midterm 2024)

**Answer:** Tryptophan

**14. What is the repeating unit of a protein?** (Biochemistry Midterm 2021)

**Answer:** Amino acids

**15. Which of the following is NOT observed at the isoelectric pH (pI) of a protein?** (Biochemistry Midterm 2021)

**Answer:** Maximum solubility **Explanation:** Lippincott states that at pI, the net charge is zero, leading to **minimum** solubility because there is no electrostatic repulsion between molecules.

**16. Carbon next to the central carbon in an amino acid is known as?** (Biochem 1st Prof 2016)

**Answer:** Beta carbon

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# II. Protein Structure & Dynamics

**17. The primary structure of protein represents?** (Biochemistry Midterm 2021)

**Answer:** Linear sequence of amino acids joined by peptide bonds

**18. In the secondary structure of protein, an amino acid that does not form an $\alpha$-helix is?** (Biochemistry Midterm 2021)

**Answer:** Proline **Explanation:** Proline is an "imino acid" with a secondary amino group; its rigid ring disrupts the $\alpha$-helix.

**19. Which is the most common secondary structure of protein?** (Biochem 2025 Mock DDC)

**Answer:** $\alpha$-helix

**20. Myoglobin is a protein characterized by which level of structure?** (Biochem 2025 Mock DDC)

**Answer:** Tertiary structure

**21. Hemoglobin is characterized by which level of structure?** (Biochem 2025 Mock DDC)

**Answer:** Quaternary structure

**22. Which of the following is responsible for specifying the 3D shape of a protein?** (Biochem 2025 Mock DDC)

**Answer:** The amino acid sequence

**23. Fibrous protein (like collagen) consists of polypeptide chains arranged in?** (Biochem 2025 Mock DDC)

**Answer:** A triple helix (specifically for collagen) or $\alpha$-helix

**24. Which of the following interactions stabilize the tertiary structure EXCEPT?** (Biochemistry Midterm 2021)

**Answer:** Peptide bonds **Explanation:** Peptide bonds are covalent bonds that define the **primary** structure; tertiary structure is maintained by non-covalent interactions (Hydrogen, Ionic, Hydrophobic).

**25. Disulphide bond is formed between which residues?** (Biochemistry Mids 2020)

**Answer:** Cysteine

**26. Which factor is responsible for the denaturation of protein?** (Biochem 2025 Mock DDC)

**Answer:** All of these (Heat, pH change, Organic solvents)

**27. Which statement about the peptide bond is incorrect?** (Biochemistry Midterm 2021)

**Answer:** Charged and polar **Explanation:** Lippincott clarifies that peptide bonds are **uncharged** but polar.

**28. How many amino acids are normally found in proteins?** (Biochem 2025 Mock DDC)

**Answer:** 20

**29. Which analytical technique separates proteins mainly based on charge?** (Biochem 2025 Mock DDC)

**Answer:** Electrophoresis

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# III. Hemoglobin & Myoglobin (Hb & Mb)

**30. Normal adult hemoglobin (HbA) has which structure?** (Biochem 2025 Mock DDC)

**Answer:** $\alpha_2\beta_2$

**31. Fetal hemoglobin (HbF) consists of which chains?** (Biochem 2025 Mock DDC)

**Answer:** $\alpha_2\gamma_2$

**32. Fetal hemoglobin (HbF) has a higher affinity for Oxygen than HbA?** (Biochem Exam 2012)

**Answer:** True

**33. The iron present in normal hemoglobin is in which oxidation state?** (Biochem 2025 Mock DDC)

**Answer:** $Fe^{2+}$ (Ferrous)

**34. Methemoglobin contains iron in which state?** (Biochem 2025 Mock DDC)

**Answer:** $Fe^{3+}$ (Ferric) **Explanation:** Ferric iron cannot bind oxygen, leading to "chocolate cyanosis".

**35. Sickle cell anemia is caused by the substitution of which amino acid?** (Biochem 2025 Mock DDC)

**Answer:** Glutamate $\rightarrow$ Valine **Explanation:** A nonpolar valine replaces a polar glutamate at position 6 of the $\beta$-globin chain.

**36. Sickle cell anemia is inherited as which trait?** (Biochem Mid Term Paper Docx)

**Answer:** Autosomal recessive

**37. Thalassemia is primarily due to?** (Biochem 2025 Mock DDC)

**Answer:** Reduced globin chain synthesis

**38. Hemoglobin S is produced by which genetic event?** (Biochem Exam 2012)

**Answer:** Point mutation at position 6 of the Beta chain

**39. Oxygen binds to hemoglobin at which specific site?** (Biochem 2025 Mock DDC)

**Answer:** Iron atom

**40. How many oxygen molecules can one hemoglobin molecule bind?** (Biochem Mid Term Paper Docx)

**Answer:** 4

**41. The oxygen dissociation curve for hemoglobin is?** (Biochem Mid Term Paper Docx)

**Answer:** Sigmoidal

**42. Myoglobin is found in which tissue?** (Biochemistry Midterm 2024)

**Answer:** Muscle tissue

**43. Which amino acids in myoglobin are likely localized in the interior?** (Biochemistry Midterm 2024)

**Answer:** Valine **Explanation:** Nonpolar/hydrophobic amino acids like Valine are buried inside the globular structure away from water.

**44. Carboxyhemoglobin is formed when hemoglobin binds to?** (Biochem 2025 Mock DDC)

**Answer:** Carbon monoxide

**45. What is the prosthetic group of hemoglobin?** (Biochem 2025 Mock DDC)

**Answer:** Heme

**46. Normal $HbA_2$ percentage in adults is?** (Biochem 2025 Mock DDC)

**Answer:** 2–3%

**47. The deoxy form of hemoglobin is also known as?** (Biochem Proff 2025)

**Answer:** T (Tense) form

**48. Patients with hemoglobin SC disease?** (Biochemistry Midterm 2021)

**Answer:** Have a different mutation in each copy of their beta-globin gene

**49. A patient homozygous for sickle cell anemia inherits which pattern?** (Biochemistry Midterm 2021)

**Answer:** SS

**50. What metal ion is found at the center of the heme group?** (Biochem 2025 Mock DDC)

**Answer:** Fe^{2+}

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